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A FLAVOUR MODEL FOR BEER FERMENTATION

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A new beer fermentation model is developed based upon fundamental knowledge of biochemical pathways. The model can be subdivided into a growth model, an amino acid model, and a flavour/aroma model. Experimentation allowed for accurate model parameter identification. The results demonstrate the capability to accurately describe batch beer fermentation dynamics.

Key Words: *Mathematical model, beer fermentation, flavour model.*

INTRODUCTION

Existing models of beer fermentation[6,8], are too simplistic to provide a good basis for process optimization. A good model of beer fermentation must include the major fermentation components (e.g. sugars, biomass, and ethanol, as well as flavour and aroma compounds which distinguish one beer from another. The model developed in this work is based on known biochemical pathways leading to the production of these flavour and aroma compounds. A model such as that of Engasser *et al.*[6] is sufficient for the purposes of on-line estimation, but does not provide the true cause and effect relationships required for predictive and optimization purposes, especially for flavour compounds.

An important consideration in model development is the obtainable measurement set. In other words, what process variables can be measured on and off-line for use in model identification? It usually does not make sense to model process variables which cannot be measured since there will be no direct feedback to verify the validity of that portion of the overall model. In some cases, it is, however necessary to model certain unmeasurable process quantities because of the role they may play in the true system dynamics. Care is necessary in these instances to avoid overparametrization.

Another consideration in modeling is process knowledge. Without adequate process knowledge, it is not possible to formulate anything more than an empirical model based solely on observations of process inputs and outputs. In the case of beer fermentation and yeast metabolism, much work has been done which provides valuable process insights for use in modeling. The fermentation model developed in this work is based on known biochemical pathways in yeast. First, a core growth model was developed describing sugar uptake, yeast growth, thermal effects, ethanol production, and carbon dioxide evolution. A nutrient model, which is dependent on the growth model, was then formulated to describe the dynamics of amino acid uptake. Finally a flavour model describing fusel alcohol, ester, and vicinal diketone dynamics was developed.

GROWTH MODEL

The models of Engasser *et al.*[6] and Gee and Ramirez[8] formed the basis of the growth model. Two new features were added. These are the modeling of carbon dioxide release and growth limitations due to availability of unsaturated fatty acids.

The basic growth model of Engasser *et al.*[6] and Gee and Ramirez[8] is as follows

Sugar Consumption

Glucose

$$\frac{dG}{dt} = -\mu_1 X \tag{1}$$

Maltose

$$\frac{dM}{dt} = -\mu_2 X \tag{2}$$

Maltotriose

$$\frac{dN}{dt} = -\mu_3 X \tag{3}$$

where the specific growth rates are

$$\mu_1 = \frac{\mu_G G}{K_G + G} \tag{4}$$

$$\mu_2 = \frac{\mu_M M}{K_M + M} \cdot \frac{K'_G}{K'_G + G} \tag{5}$$

$$\mu_3 = \frac{\mu_N N}{K_N + N} \cdot \frac{K'_G}{K'_G + G} \cdot \frac{K'_M}{K'_M + M} \tag{6}$$

and the temperature dependency on the specific growth rates are given by

$$\mu_i = \mu_{i0} \exp\left[-E\mu_i/RT^2\right] \quad , \quad i = G, M, N \tag{7}$$

$$K_i = K_{i0} \exp\left[-E_{K_i}/RT^2\right] \quad , \quad i = G, M, N \tag{8}$$

$$K'_i = K'_{i0} \exp\left[-E'_{K_i}/RT^2\right] \quad , \quad i = G, M \tag{9}$$

Biomass production

$$\frac{dX}{dt} = \mu_x X \tag{10}$$

where

$$\mu_x = [Y_{XG}\mu_1 + Y_{XM}\mu_2 + Y_{XN}\mu_3] \tag{11}$$

Ethanol production

$$E = E_0 + Y_{EG}(G_0 - G) + Y_{EM}(M_0 - M) + Y_{EN}(N_0 - N) \tag{12}$$

Temperature

$$\frac{dT}{dt} = \frac{1}{\rho C_p}\left[-X\left(\Delta H_{FG}\mu_1 + \Delta H_{FM}\mu_2 + \Delta H_{FN}\mu_3\right) - u(T - Tc)\right] \tag{13}$$

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In this model, yeast growth is considered to depend solely on the uptake of the three sugars present in the media. Actually, yeast growth has been found to be limited by the availability of unsaturated fatty acids and lipids in the wort[13,15]. The process of cell division can only continue as long as the necessary structural components can be obtained from the media or synthesized within the cells. Unsaturated fatty acids used in membrane structures cannot be synthesized by the cell in the absence of oxygen. Therefore, in anaerobic fermentation, the available unsaturated fatty acids become depleted. Engasser *et al.*[6] observed this cell growth limitation and described it by using a variable biomass yield which decreased with the increasing extent of fermentation as evidenced by increases in the ethanol concentration. However, an ethanol concentration increase is not really physically representative of the cause of this growth limitation. A more physiologically accurate model would consider cell growth as limiting itself. The most obvious way to include this effect in the model is to assume a simple feedback inhibition mechanism for cell growth as

$$\mu_x = [Y_{XG}\mu_1 + Y_{XM}\mu_2 + Y_{XN}\mu_3]\frac{K_x}{K_x + (X - X_0)^n} \quad (14)$$

Typically $n = 2$ was used for this work.

In modeling the release of carbon dioxide we consider that the cells produce carbon dioxide as a gas. This gas is absorbed into the liquid phase though a mass transfer process described by the mass transfer coefficient K_{GL}. There is, of course, a saturation limit given as C_{sat}. The carbon dioxide generation is therefore modelled as

Liquid phase

$$\frac{dC_l}{dt} = \begin{cases} K_{GL}(C_{sat} - C_l) & \text{for } C_l < C_{sat} \\ 0 & \text{for } C_l = C_{sat} \end{cases} \quad (15)$$

Gas phase

$$\frac{dC_g}{dt} = \begin{cases} (Y_{CG}\,\mu_1 + Y_{CM}\,\mu_2 + Y_{CN}\,\mu_3)X \\ \quad - K_{GL}(C_{sat} - C_l) & \text{for } C_l < C_{sat} \\ (Y_{CG}\,\mu_1 + Y_{CM}\,\mu_2 + Y_{CN}\,\mu_3)X & \text{for } C_l = C_{sat} \end{cases} \quad (16)$$

Known model parameters from Gee and Ramirez[8] and Gee[7] are given in Table I.

Amino acids

Certain amino acids have been tied to the formation of specific flavour compounds in beer, particularly the fusel alcohols[1,5,16]. Therefore, it is important to include the dynamics of amino acid uptake in the model. Amino acids are utilized by the cell primarily as structural components in the construction of structural and functional proteins. The higher the rate of growth, the greater the cell's need for amino acids. The rate of amino acid assimilation by yeast is therefore modeled as negatively proportional to the yeast growth rate. However, this rate is limited by the availability of amino acids in the media. The rate expression used is

$$\frac{d[AA]}{dt} = -Y_{AA/X}\frac{[AA]}{K_{AA} + [AA]}\frac{dX}{dt} \quad (17)$$

TABLE I Known Model Parameters

Sugar	Y_{Ei}	Y_{Xi}	Y_{Ci}	ΔH_{Fi}, kJ/mol
Glucose	1.92	.134	1.97	− 91.2
Maltose	3.84	.268	3.94	−226.3
Maltotriose	5.76	.402	5.91	−361.3

$K_{GL} = 0.07\ \text{hr}^{-1}$

Another phenomenon that was observed with the uptake of amino acids was a lag phase at the beginning of fermentation. The exact cause of this delay is not readily available. One possible cause could be that the enzymes needed to transport amino acids across the cellular membranes may not have been present in any significant quantity within the cell as the outset of fermentation. This seems reasonable considering that the cells were kept in a relatively nutrient-free environment under non-growth conditions prior to inoculation. Since the delay time was relatively constant for each data set, we modeled it as a first order exponential decay whose time constant was the same for all fermentations.

The three amino acids of concern are leucine (L), isoleucine (I), and valine (V). The model equations for these amino acids are

$$\frac{dL}{dt} = -Y_{LX}\frac{dX}{dt}\frac{L}{K_L + L}D \quad (18)$$

$$\frac{dI}{dt} = -Y_{IX}\frac{dX}{dt}\frac{I}{K_I + I}D \quad (19)$$

$$\frac{dV}{dt} = -Y_{VX}\frac{dX}{dt}\frac{V}{K_V + V}D \quad (20)$$

where

$$D = 1 - e^{-t/\tau_d} \quad (21)$$

Flavour compounds

Most industrial fermentation processes are designed to produce a single desired product in quantity which must then be separated from the media and purified. In beer fermentation, the brewing media at the end of the process time is itself the final product. This makes beer an extremely complex product with respect to chemical composition. In this work, beer flavour is characterized with 10 chemical species. These compounds fall into three categories: fusel alcohols, esters, and vicinal diketones. The modeling of each of these groups is considered as follows.

Fusel alcohols

Fusel alcohols contribute a plastic, solvent-like flavour to beer. This is an undesirable flavour characteristic. Another equally undesirable effect is a strong contribution to the physiological symptoms comprising a hangover. The fusel alcohols are also involved in the biochemical pathways leading to the formation of esters, another important flavour group. Four fusel alcohols are considered in our model: n-propanol, isobutyl alcohol, isoamyl alcohol, and 2-methyl-1-butanol.

Äyräpää[2] has given the scheme for the formation of fusel alcohols in yeast fermentation. Two major biochemical pathways lead to the formation of fusel alcohols. One is the synthetic pathway and the other the Ehrlich mechanism. Both proceed by way of the intracellular oxo acid pool. The synthetic pathway depends on the carbohydrate metabolism for the precursor molecules needed for oxo acid synthesis. In the Ehrlich mechanism, exogenous amino acids are transported into the cell and transaminated by oxoglutarate to form their corresponding oxo acids. The oxo acids produced in either of these pathways can then be decarboxylated and reduced by cellular enzymes to form fusel alcohols which are then secreted into the media.

The extracellular factors influencing the rate of fusel alcohol formation are the sugar and amino acid uptake rates. At high sugar uptake rates, large amounts of metabolic precursors are available inside the cell for synethesizing oxo acids. Similarly, when amino acid uptake rates are high, the intracellular oxo acid pool will swell, giving rise to higher rates of fusel alcohol formation. However, the availability of

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precursor molecules is not the regulating concern in the synthetic pathway. The operation of this pathway is instead governed by cell growth rate. At low growth rates, fewer oxo acids and amino acids are required within the cell, so the synthetic pathway will be less active. The opposite is true at high growth rate conditions. There is also a cellular regulating mechanism which controls the activity of the two pathways with relation to each other. Äyräpää and Palmqvist[3] found that the synthetic pathway was completely halted at high amino acid concentrations, while at low amino acid concentrations, the synthetic pathway was found to be the primary means of fusel alcohol formation. Based upon this process knowledge, the following relationship is used to describe the general rate of fusel alcohol production.

$$\frac{d[OH_j]}{dt} = Y_{OH_j/S}\mu_x X \frac{K_{I,AA_j}}{K_{I,AA_j}+[AA_j]} + Y_{OH_j/E}\mu_{AA}X \quad (22)$$

In this expression, OH_j denotes the alcohol derived from the oxo acid which corresponds to amino acid AA_j. The first term models the synthetic pathway growth rate mechanism that is inhibited by high amino acid concentrations. The second term represents the Ehrlich mechanism which is proportional to the corresponding amino acid uptake rate.

Of the four fusel alcohols considered, three could easily be traced to the amino acids from which they are derived. Isobutyl alcohol arises from the decarboxylation and subsequent reduction of the oxo acid corresponding to L-valine. Similarly, isoamyl alcohol and 2-methyl-1-butanol are derived from the amino acids L-leucine and L-isoleucine, respectively. The equations describing the formation of these three fusel alcohols are therefore

$$\frac{dIB}{dt} = Y_{IB/S}\,\mu_x X \frac{K_{I,V}}{K_{I,V}+V} + Y_{IB/E}\,\mu_V X \quad (23)$$

$$\frac{dIA}{dt} = Y_{IA/S}\,\mu_x X \frac{K_{I,L}}{K_{I,L}+L} + Y_{IA/E}\,\mu_L X \quad (24)$$

$$\frac{dMB}{dt} = Y_{MB/S}\,\mu_x X \frac{K_{I,I}}{K_{I,I}+I} + Y_{MB/E}X \quad (25)$$

where

$$\mu_V = -\frac{1}{X}\frac{dV}{dt} \quad (26)$$

$$\mu_L = -\frac{1}{X}\frac{dL}{dt} \quad (27)$$

$$\mu_I = -\frac{1}{X}\frac{dI}{dt} \quad (28)$$

The origin of n-propanol is not so clearly defined. It seems to arise from the reduction of the acid propionate. The reduction of intracellular acids involved in the fatty acid metabolism to form alcohols has been observed[12]. In these metabolic pathways, the acids under consideration exist in a complexed form with coenzyme A. Therefore, the reduction of propionyl-CoA is the most likely pathway to the formation of n-propanol. Besides being an end product of the intracellular catabolism of odd-chain fatty acids, propionyl-CoA is also formed in the biological pathways involved in breaking down the amino acids valine and isoleucine. Since these two amino acids can exist in the cell as the result of either the synthetic pathway or membrane transport, a total of five different pathways are available for the production of n-propanol.

The terms in the n-propanol formation model relating to the synthetic and amino acid transport pathways should logically parallel the expressions in Equation 22. The other pathway leading to n-propanol must result from the uptake and catabolism of odd-chain fatty acids from the wort since odd-chain fatty acids are not produced in the cellular metabolism. This uptake was assumed to be directly related to the yeast growth rate. Therefore, the model describing the formation of n-propanol is

$$\frac{dP}{dt} = Y_{PV/S}\,\mu_x X\frac{K_{IV}}{K_{IV}+V} + Y_{PV/E}\,\mu_V X + Y_{PI/S}\mu_x X\frac{K_{II}}{K_{II}+I} + Y_{PI/E}\,\mu_I X + Y_{P/X}\,\mu_x X \quad (29)$$

Esters

Esters are one of the more volatile compounds in beer and hence, contribute a great deal to beer aroma. In moderate quantities, they can add a pleasant, full-bodied character to beer aroma. When present in excess, however, they give beer aroma an overly fruity quality which is considered undesirable by most consumers. Within the cell, an acyl-CoA compound reacts with an alcohol in the presence of the proper enzyme to form an ester. Acyl-CoA compounds arise in such pathways as the fatty acid synthesis and breakdown pathways, and the glycolysis pathway. This cellular esterfication pathway was elucidated by Nordström[14].

Three esters were included in our model and are ethyl acetate, ethyl caproate, and isoamyl acetate. While each arises from the intracellular pathway discussed above, each had to be modeled differently according to the individual acids and alcohols involved. The most straightforward ester to model is ethyl acetate which arises from the enzymatic esterfication of ethanol and acetyl-CoA. Because both of these compounds occur in the cell as the result of sugar metabolism, their intracellular concentration, and thus availability for esterfication, will be greater at high sugar uptake rates. It is therefore reasonable to assume that the ethyl acetate production rate should negatively parallel the uptake rate of the sugars.

$$\frac{d[EA]}{dt} = Y_{EA/S}[\mu_1+\mu_2+\mu_3]X \quad (30)$$

Ethyl caproate is generated in the cell as the result of a reaction between ethanol and caproyl-CoA. Caproyl-CoA exists as an intermediate product in the fatty acid anabolic and catabolic pathways. As such, it should be present in relatively small quantities in comparison with ethanol which is the end product of the major cellular pathway operating under anaerobic conditions of fermentation. Therefore, the availability of caproyl-CoA can be considered as rate limiting. The activity of the fatty acid synthesis and breakdown pathways is directly related to the cell growth. The ethyl caproate model used is therefore

$$\frac{d[EC]}{dt} = Y_{EC/X}\,\mu_x X \quad (31)$$

The third ester considered is isoamyl acetate which results from the reaction of isoamyl alcohol with acetyl-CoA inside the cell. Acetyl-CoA should be present in the cell in great excess compared with isoamyl alcohol which is a minor metabolite. The availability of isoamyl alcohol can be considered rate limiting. Therefore, a first order isoamyl acetate production rate model is

$$\frac{d[IAc]}{dt} = Y_{IAc}\,\mu_{IA}X \quad (32)$$

Vicinal Diketones (VDK)

The vicinal diketones produced in beer fermentation are considered very undesirable flavour compounds. They contribute a buttery flavour note which is not pleasant in beer. The two primary VDKs are diacetyl (butanedione) and 2,3-

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pentanedione. VDKs are produced early in fermentation and are later reassimilated by the yeast. A compound that exhibits similar dynamic behaviour is acetaldehyde. This compound is also considered to have a derogatory effect on flavour and was grouped in the model with the VDKs.

The biochemical pathways leading to the formation of VDKs have been studied by Geiger and Piendl[10]. The main precursors are acetohydroxy acids which are found in the pathway of pyruvate conversion to the various oxo acids. Since the products of this pathway are the oxo acids which are used for growth, we model the formation of VDKs as proportional to the growth rate. The reassimilation of VDKs by yeast is assumed proportional to the VDK concentration. Therefore, the final VDK model is

$$\frac{d[VDK]}{dt} = Y_{VDK}\mu_x X - k_{VDK}[VDK]X \qquad (33)$$

The brewer is not usually as concerned with the concentrations of the individual VDKs as with the total VDK composition. Since the dynamics of diacetyl and 2,3-pentanedione parallel each other, we lump these two compounds into a single variable called VDK.

Acetaldehyde exhibits similar dynamics to that of VDK in that it too is produced early in fermentation and later reassimilated. Geiger and Piendl[11] have studied the pathways resulting in acetaldehyde formation in yeast. The primary source of acetaldehyde is the partial reduction of acetate resulting from acetyl-CoA. This acetyl-CoA is found in both the fatty acid pathways (related to growth) and in the glycolysis pathway (related to sugar uptake). Acetaldehyde itself is the direct precursor to ethanol at the end of the glycolysis pathway, but is only the product of a minor side reaction with relation to the fatty acid metabolic processes. For this reason, the production of acetaldehyde should be considered as related primarily to the sugar uptake rates rather than the cell growth rate. The model for acetaldehyde dynamics is

$$\frac{d[AAl]}{dt} = Y_{AAl}[\mu_1 + \mu_2 + \mu_3]X - k_{AAl}[AAl]X \qquad (34)$$

Parameter Identification

Given the data set for a typical batch fermentation, there were certain considerations which had to be made concerning the identifiability of the model parameters. In this context, identifiability refers to the ability of the data set to uniquely determine the parameter set which best fits the data and yields a realistic model representation of the physical phenomena involved. With this view of identifiability in mind, the model was reviewed and altered where necessary.

The measurements available to fit the growth portion of the model were the three sugar concentrations, the temperature, the ethanol concentration, and the carbon dioxide evolution rate. Based solely on these measurements, the inhibition parameter, K_x, in the biomass equation could not be determined. This inhibition was, however, reflected in the amino acid uptake data as seen in Figure 3. Here the amino acids were not completely consumed by the yeast, but their uptakes were limited at the end of the real growth phase. Therefore, the data set with the amino acids provided the necessary feedback to allow the parameter, K_x, to be identified.

The fusel alcohol model presented the first real problem in identifiability. The measurements available here were those of the four fusel alcohols and three amino acids along with the growth data. These data did not provide any way to discern between the synthetic and Ehrlich pathways. In order to discern the selectivity between these pathways, experiments using media lacking key amino acids would be necessary. The experimental data showed that all key amino acids were present throughout typical fermentations (Figure 3). Also, the individual fusel alcohols formed were seen to correlate well with the uptake of the corresponding amino acids (Figures 4 to 7). These observations indicate that under the conditions of our experiments, the synthetic pathway is negligible. The model equations for isobutyl alcohol, isoamyl alcohol, and 2-methyl-1-butanol were reduced to

$$\frac{d[IB]}{dt} = Y_{IB/E}\mu_V X \qquad (35)$$

$$\frac{d[IA]}{dt} = Y_{IA/E}\mu_L X \qquad (36)$$

$$\frac{d[MB]}{dt} = Y_{MB/E}\mu_I X \qquad (37)$$

The same arguments also hold for the synthetic pathways for the production of n-propanol. This still leaves three distinct pathways between which the data could not distinguish. It is assumed that the pathway of odd-chain fatty acid catabolism comprises a relatively minor fraction of the overall cellular metabolism; therefore, this pathway to n-propanol would be negligible compared to the pathways related to valine and isoleucine. The effects of valine and isoleucine uptake have been combined into a single uniquely identifiable term. These alterations led to the following modified model for n-propanol

$$\frac{d[P]}{dt} = Y_{P/E}[\mu_V + \mu_I]X \qquad (38)$$

Results

Three batch experiments were conducted to obtain data needed to fit the proposed beer fermentation model. A different wort was used for each experiment, and constant temperature data were obtained every 12 hours for fermentation at 10.5°C, 12°C and 14.5°C. The data obtained consisted of the initial cell concentration, temperature, carbon dioxide evolution rate, and concentration histories for glucose, maltose, maltotriose, ethanol, L-leucine, L-valine, L-isoleucine, isobutanol, isoamyl acetate, acetaldehyde, diacetyl, and 2,3-pentanedione.

Rather than fitting the entire model parameter set at one time, it was deemed more efficient to perform the fit in pieces. The rationale behind this included minimizing the computational burden while allowing the experimenter to examine each portion of the model independently to assess how well the model structure was able to describe the observed phenomena. The computational burden in parameter identification increases with the square of the number of parameters being determined. The opportunity for the experimenter to focus attention on each portion of the model independently provides a basis for improving portions of the model structure if necessary.

The model was broken into four parts which could be fit independently when approached sequentially. First, the growth and amino acid models were combined and fitted to the data. This yielded 16 unknown parameters. The growth and amino acid equations are fully independent of the flavour models. Once this fit had been completed, the fusel alcohol and VDK models, which depend on the growth and amino acid models but not on each other, were independently fit to the data. This procedure yielded the values of the four parameters contained in each of these models. Finally, using the results of the growth, amino acid, and fusel alcohol models, the three parameters in the ester model were determined.

The algorithm used to fit the model parameters was developed by Dennis and Schnabel[4], and is based on a robust scheme for unconstrained numerical minimization of non-

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TABLE II Model Parameters and Confidence Intervals

Parameter	At 10.5°C Value	At 12°C Value	At 14.5°C Value
μ_G	$0.01348 \pm 1.014 \times 10^{-5}$	$0.01606 \pm 2.008 \times 10^{-7}$	$0.02554 \pm 7.569 \times 10^{-8}$
μ_M	$0.02581 \pm 1.203 \times 10^{-5}$	$0.02563 \pm 6.276 \times 10^{-6}$	$0.02266 \times 1.138 \times 10^{-7}$
μ_N	$0.09881 \pm 6.073 \times 10^{-5}$	$0.07069 \pm 2.998 \times 10^{-4}$	$0.03913 \pm 3.590 \times 10^{-6}$
K_G	$0.7464 \pm 7.371 \times 10^{-5}$	$0.0000 \pm 8.791 \times 10^{-5}$	$0.0000 \pm 1.188 \times 10^{-6}$
K_M	$40.97 \pm 3.984 \times 10^{-4}$	$45.11 \pm 3.700 \times 10^{-2}$	$125.2 \pm 1.900 \times 10^{-3}$
K_N	$250.0 \pm 8.130 \times 10^{-3}$	$5000. \pm 2.151 \times 10^{1}$	$3000. \pm 2.814 \times 10^{-1}$
K'_G	$5.356 \pm 5.071 \times 10^{-3}$	$41.96 \pm 3.380 \times 10^{-3}$	$141.3 \pm 2.000 \times 10^{-3}$
K'_M	$13.17 \pm 2.127 \times 10^{-5}$	$1 \times 10^{6} \pm 1.12 \times 10^{7}$	$1 \times 10^{6} \pm 4.48 \times 10^{5}$
K_x	$365000 \pm 1.989 \times 10^{-3}$	$365000 \pm 6.111 \times 10^{2}$	$365000 \pm 8.888 \times 10^{2}$
Y_{LX}	$0.07734 \pm 4.689 \times 10^{-2}$	$0.09286 \pm 2.900 \times 10^{-3}$	$0.07953 \pm 2.277 \times 10^{-6}$
Y_{IX}	$0.02171 \pm 1.883 \times 10^{-5}$	$0.06608 \pm 3.400 \times 10^{-3}$	$0.02119 \pm 7.056 \times 10^{-6}$
Y_{VX}	$0.02045 \pm 4.144 \times 10^{-5}$	$0.04088 \pm 2.000 \times 10^{-3}$	$0.02048 \pm 1.159 \times 10^{-5}$
K_L	$0.5905 \pm 2.717 \times 10^{-5}$	$0.8449 \pm 6.900 \times 10^{-3}$	$0.7156 \pm 4.840 \times 10^{-5}$
K_I	$0.07191 \pm 3.487 \times 10^{-5}$	$0.9496 \pm 2.090 \times 10^{-2}$	$0.07288 \pm 1.620 \times 10^{-4}$
K_L	$0.02769 \pm 1.179 \times 10^{-4}$	$0.0067 \pm 4.300 \times 10^{-2}$	$6.566 - 5.092 \times 10^{-4}$
τ_d	23.54	$23.54 \pm 1.472 \times 10^{1}$	23.54
$Y_{IB/E}$	$0.1607 \pm 1.800 \times 10^{-3}$	$0.2405 \pm 3.800 \times 10^{-3}$	$0.2034 \pm 9.509 \times 10^{-4}$
$Y_{IA/E}$	$0.5128 \pm 9.726 \times 10^{-4}$	$0.5631 \pm 1.700 \times 10^{-3}$	$0.5958 \pm 5.077 \times 10^{-4}$
$Y_{MB/E}$	$0.3840 \pm 2.100 \times 10^{-3}$	$0.5166 \pm 3.700 \times 10^{-3}$	$0.5154 \pm 1.100 \times 10^{-3}$
$Y_{P/E}$	$0.2216 \pm 9.585 \times 10^{-4}$	$0.2448 \pm 1.900 \times 10^{-3}$	$0.2395 \pm 5.107 \times 10^{-4}$
$Y_{EA/S}$	$7.520 \times 10^{-4} \pm 3.753 \times 10^{-6}$	$9.870 \times 10^{-4} \pm 5.973 \times 10^{-6}$	$1.237 \times 10^{-3} \pm 2.950 \times 10^{-6}$
$Y_{EC/X}$	$1.260 \times 10^{-4} \pm 3.124 \times 10^{-6}$	$1.220 \times 10^{-4} \pm 4.792 \times 10^{-6}$	$1.066 \times 10^{-4} \pm 2.580 \times 10^{-6}$
Y_{IAc}	$2.918 \times 10^{-2} \pm 1.896 \times 10^{-4}$	$2.331 \times 10^{-2} \pm 2.921 \times 10^{-4}$	$2.812 \times 10^{-2} \pm 1.468 \times 10^{-4}$
Y_{VDK}	$6.730 \times 10^{-5} \pm 3.771 \times 10^{-5}$	$1.040 \times 10^{-4} \pm 2.931 \times 10^{-4}$	$1.444 \times 10^{-4} \pm 4.992 \times 10^{-5}$
k_{VDK}	$1.818 \times 10^{-5} \pm 4.871 \times 10^{-5}$	$3.980 \times 10^{-6} \pm 5.973 \times 10^{-6}$	$5.116 \times 10^{-5} \pm 5.648 \times 10^{-5}$
Y_{AAl}	$3.889 \times 10^{-3} \pm 4.575 \times 10^{-7}$	$6.467 \times 10^{-3} \pm 4.792 \times 10^{-6}$	$5.278 \times 10^{-3} \pm 3.013 \times 10^{-5}$
k_{AAl}	$3.914 \times 10^{-5} \pm 1.016 \times 10^{-6}$	$1.734 \times 10^{-4} \pm 1.320 \times 10^{-4}$	$6.339 \times 10^{-5} \pm 1.042 \times 10^{-6}$

linear functions. The routine used a quasi-Newton search technique based on numerical estimates of the system gradient and Hessian to search out the parameter space for the optimum. Since the scheme could converge on a local minimum rather than the global minimum, it was necessary to try a range of different starting values for the parameters. Although the routine was designed for unconstrained minimization, it is possible to include constraints by posing them in the form of penalty functions added to the overall objective function. In the proposed model, parameters had to be constrained to be non-negative.

The basic objective function used was the sum of squares of the errors between the model predictions and the available data points. Because of the large discrepancies between the orders of magnitude of the process variables, normalization constants were used to scale the various quadratic terms in the objective function.

The parameter values determined by the fitting procedure are given in Table II along with their 95% confidence limits. In general, each model fit was good with reasonable confidence intervals associated with most of the parameter values, good visual fits of the data, and physiologically realistic parameter values. A few of the parameters were consistently insensitive to the data in each of the experiments. The most obvious of these is K_x which took on the same value in all three cases. The form of the empirical yeast inhibition term caused this parameter to be fairly insensitive in the model fitting process. It does make sense, however, for this parameter to be independent of temperature since it is influenced only by the unsaturated fatty acids in the wort. The inoculum size was essentially identical in each experiment as was aeration. Therefore, the growth limitations should be similar in each run. As seen by the confidence limits determined for K_x, its value is highly reliable in each fit.

The most insensitive parameter in these experiments was the inhibition constant for maltose at the two higher temperatures. This parameter is only important in the region where maltose inhibition begins to lessen and the rate of maltotriose uptake increases. The only experiment in which this phenomenon was observed was that performed at 10.5°C. For this run, the fit was excellent. In the two runs exhibiting insensitivity, this parameter was not needed in the model structure because maltotriose was never taken up significantly.

In running experiments at three different temperatures with the Arrhenius temperature dependency on the kinetic parameters, it had been hoped to obtain unique kinetic parameters for all three runs as obtained in our prior work[8]. In the prior work, fermentations were carried out simultaneously using identical media and different fermentors. In this work, the runs were made sequentially using different worts but the same highly instrumented fermentor. As can be seen by run-to-run variations in the growth kinetic parameters, variations in the media strongly affect the kinetic growth parameters in addition to temperature effects. These media differences prevented determining the unique temperature dependency on the reaction rate parameters. This fact points out the need for an adaptive model to identify growth parameters for any control or estimation application connected with beer fermentation. Such an adaptive approach is presented by Gee and Ramirez[9].

The sugar model fits are quite accurate as shown in Figures 1 and 2. Figure 1 gives the fit for the 10.5°C run where maltotriose uptake is significant. Figure 2 is at 14°C and illustrates a case where maltotriose uptake was not significant. These results demonstrate the ability of the model structure used to ably describe the uptake of sugars in beer fermentation.

The amino acid model fits also describe the data well as illustrated in Figure 3 for the 10.5°C run. The uptake of the three amino acids followed a similar pattern in experiments at all three temperatures. The effect of the yeast growth inhibition term on amino acid uptake is shown by the tail-off of the uptake curves in each case with significant amounts of each amino acid remaining in the media.

Typical fusel alcohol fits are given in Figures 4 to 7 for the run at 12°C. The model was able to describe the data fairly well. The data for all of the alcohols except isobutanol were unusual at the lowest experimental temperature. The production rates for these fusel alcohols are expected to tail-off as shown in Figures 4 to 7 as the uptake rates of the amino

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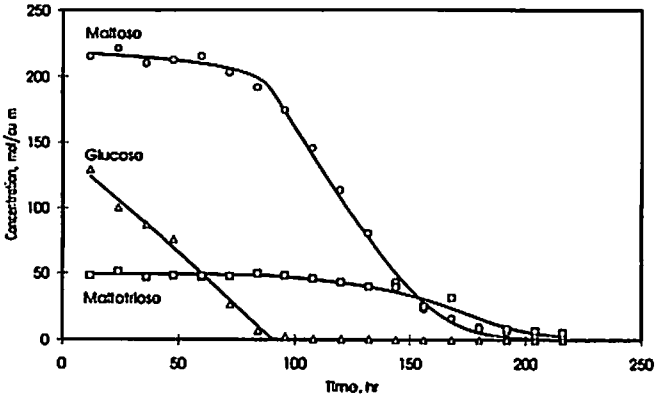


Figure 1. Model fits to sugar data at 10.5°C. (△) glucose data, (○) maltose data, (□) maltotriose data, (——) model fits.

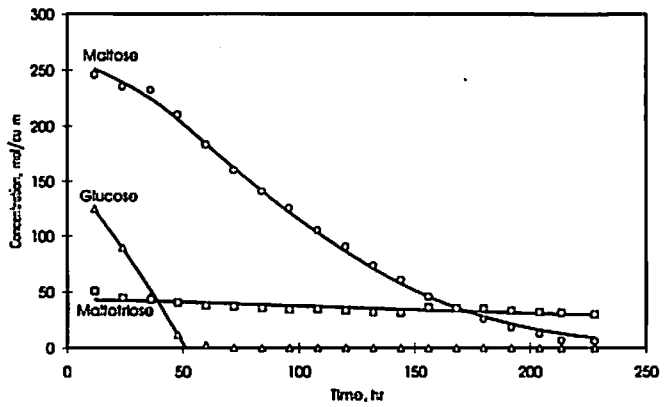


Figure 2. Model fits to sugar data at 14.5°C. (△) glucose data, (○) maltose data, (□) maltotriose data, (——) model fits.

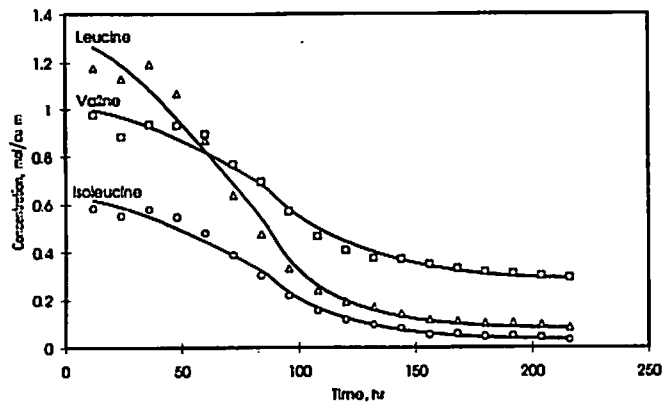


Figure 3. Model fits to amino acid data at 10.5°C. (△) leucine data, (○) isoleucine data, (□) valine data, (——) model fits.

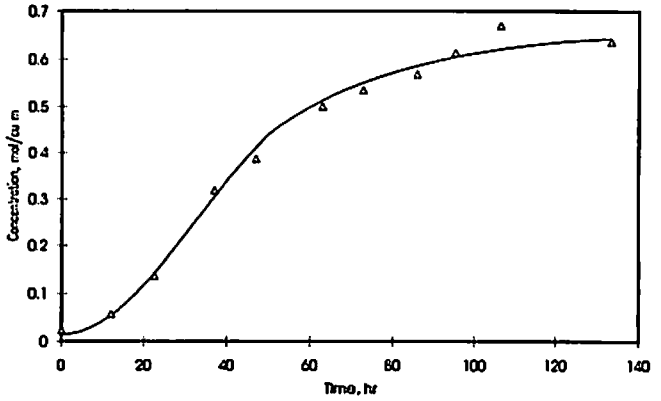


Figure 4. Model fit to isoamyl alcohol data at 12°C. (△) data, (——) model fit.

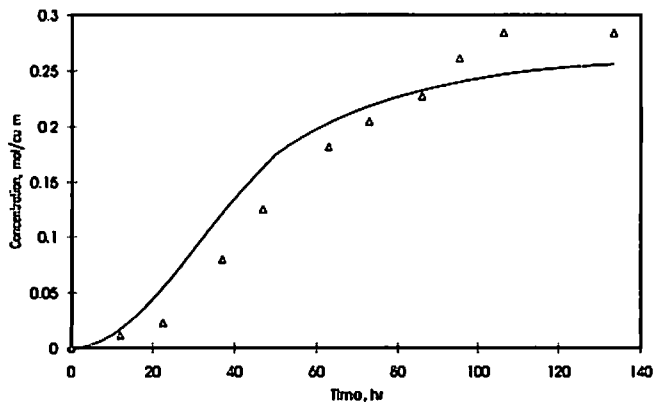


Figure 5. Model fit to 2-methyl-1-butanol data at 12°C. (△) data, (——) model fit.

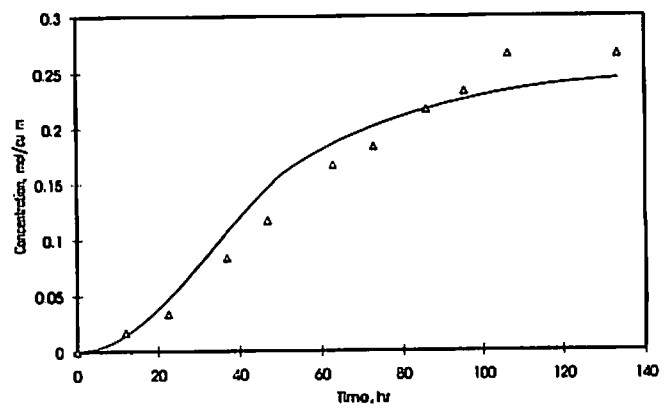


Figure 6. Model fit to n-propanol data at 12°C. (△) data, (——) model fit.

acids decrease due to growth limitation late in fermentation. For some unknown reason, this was not observed at 10.5°C with three of the four fusel alcohols. The reason is unclear. Judging from the data accumulated in the other two experiments, it seems safe to assume that the 10.5°C fusel alcohol data could be anomalous.

The ester model fits are given in Figures 8 to 10 for the 10.5°C run. The data relating to the production of isoamyl acetate are fairly poor as shown in Figure 8. This compound was present in such small amounts that, apparently, it could be quantified only at discrete values over its usual concentration range in beer. The reason for this probably lies in the fact that it is coanalyzed with ethyl acetate which appears in such relative abundance that instrument sensitivity to isoamyl acetate is very low. Also a definite lag is seen in the ester data which is not considered in the present model. These delays

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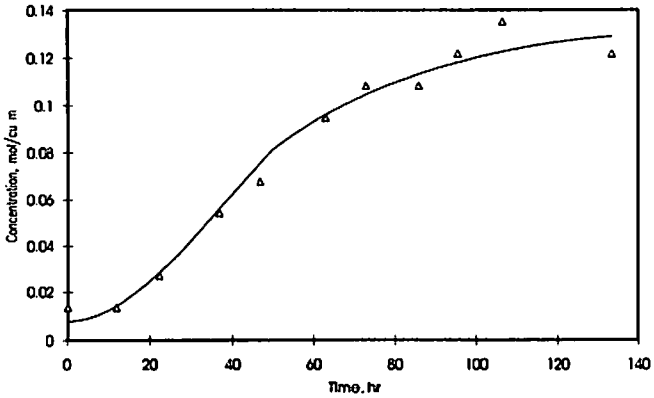


Figure 7. Model fit to isobutanol data at 12°C. (△) data, (——) model fit.

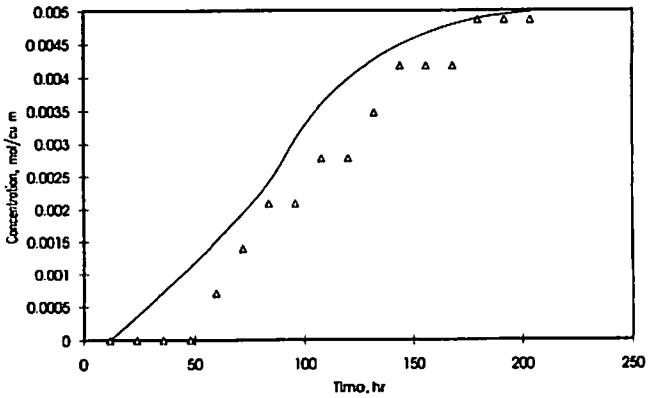


Figure 10. Model fit to ethyl caproate data at 10.5°C. (△) data, (——) model fit.

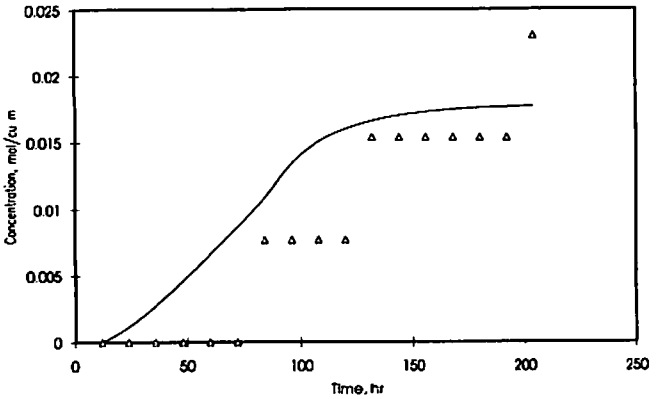


Figure 8. Model fit to isoamyl acetate data at 10.5°C. (△) data, (——) model fit.

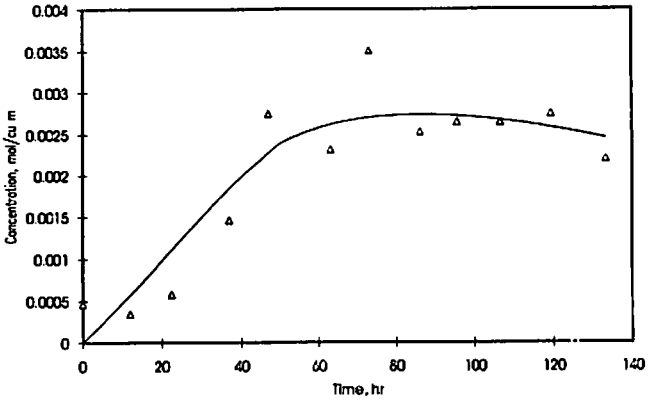


Figure 11. Model fit to VDK data at 12°C. (△) data, (——) model fit.

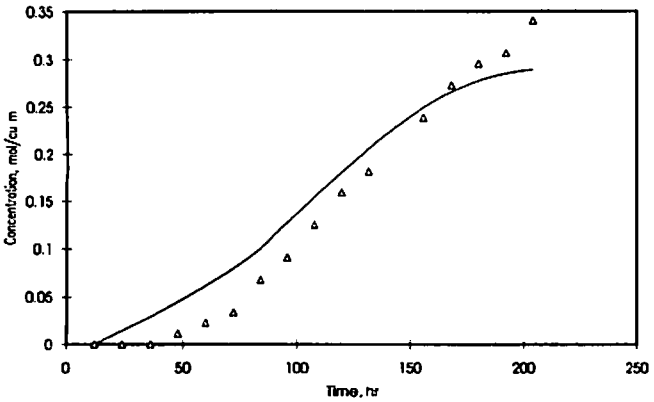


Figure 9. Model fit to ethyl acetate data at 10.5°C. (△) data, (——) model fit.

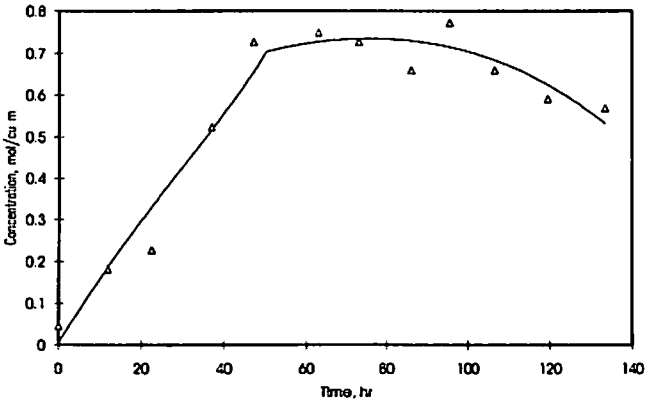


Figure 12. Model fit to acetaldehyde data at 12°C. (△) data, (——) model fit.

did appear to be uniform and could easily be modeled by including first order delay terms in the dynamic equations for each ester. This was not done in this work, but should be considered in future developments of the model.

In general, the data for the VDK concentration exhibited a greater amount of scatter than the other data obtained in this study (Figure 11). This was probably due to the fact that this compound is present in such small quantities that its analysis is extremely sensitive to signal noise levels. The acetaldehyde data also showed a reasonably large amount of scatter (Figure 12). This may be due to separation difficulties often encountered in the analysis of aldehydes in gas chromatography. Plots of the model fits for these compounds are shown in Figures 11 and 12 for the run at 12°C. As shown in these figures, despite data scatter, the model was able to describe the formation and uptake of these compounds quite

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well. It is especially impressive how well the model was able to fit the acetaldehyde data (Figure 12). The uptake term was able to take over and dominate the dynamics so well as to seemingly contribute a discontinuity to the concentration profile.

Conclusions

This study developed a new beer fermentation model based upon known biochemical pathways. The model can be subdivided into a growth model, an amino acid model, and a flavour/aroma model. In general, the model fit to experimental data was good with reasonable confidence limits associated with model parameters.

Based upon the results of this study, the beer fermentation model developed demonstrates the capability to accurately describe the uptake of sugars and amino acids as well as the production of various flavour and aroma compounds associated with beer fermentation. The model is deemed adequate for use in on-line estimation and control studies, as well as for process optimization purposes.

Nomenclature

$[AA_j]$	amino acid concentration, mol/m^3
$[AAl]$	acetaldehyde concentration, mol/m^3
C_g	CO_2 gas phase concentration, mol/m^3
C_l	CO_2 aqueous phase concentration, mol/m^3
C_p	mean heat capacity of wort, J/kg K
C_{sat}	CO_2 saturation concentration, mol/m^3
D	first order time delay
ΔH_{FG}	overall heat of fermentation for glucose, J/mol
ΔH_{Fi}	overall heat of fermentation for sugar i, J/mol
ΔH_{FM}	overall heat of fermentation for maltose, J/mol
ΔH_{FN}	overall heat of fermentation for maltotriose, J/mol
E	ethanol concentration, mol/m^3
E_{Ki}	Arrhenius activation energy for K_i, cal/mol
E_{Ki}'	Arrhenius activation energy for K'_i, cal/mol
E_{μ_i}	Arrenhuis activation energy for μ_i, cal/mol
$[EA]$	ethyl acetate concentration, mol/m^3
$[EC]$	ethyl caproate concentration, mol/m^3
$[FA]$	fatty acid concentration, mol/m^3
G	glucose concentration, mol/m^3
I	isoleucine concentration, mol/m^3
IA	isoamyl alcohol concentration, mol/m^3
$[IAc]$	isoamyl acetate concentration, mol/m^3
IB	isobutyl alcohol concentration, mol/m^3
K_a	Michaelis constant for acyl-CoA, mol/m^3
K_{AA}	Michaelis constant for amino acid uptake, mol/m^3
k_{AAl}	effective first-order rate constant for uptake of AAl, m^3/mol hr
K_G	Michaelis constant for glucose, mol/m^3
K'_G	inhibition constant for glucose, mol/m^3
K_{GL}	gas-liquid mass transfer coefficient, hr^{-1}
K_{i0}	Arrhenuis frequency factor for Ki, mol/m^3
K'_{i0}	Arrhenuis frequency factor for $K'i$, mol/m^3
K_I	Michaelis constant for isoleucine, mol/m^3
$K_{I,AAj}$	inhibition constant for amino acid AA_j, mol/m^3
$K_{I,I}$	inhibition constant for isoleucine, mol/m^3
$K_{I,L}$	inhibition constant for leucine, mol/m^3
$K_{I,V}$	inhibition constant for valine, mol/m^3
K_L	Michaelis constant for leucine, mol/m^3
K_M	Michaelis constant for maltose, mol/m^3
K'_M	inhibition constant for maltose, mol/m^3
K_N	Michaelis constant for maltotriose, mol/m^3
K_V	Michaelis constant for valine, mol/m^3
k_{VDK}	effective first-order rate constant for uptake of VDK, m^3/mol/hr
K_{VDK}	Michaelis constant for VDK uptake, mol/m^3
K'_x	yeast growth inhibition constant, mol/m^3
K_x	empirical yeast growth inhibition constant, (mol/m^3)^2
L	leucine concentration, mol/m^3
MB	2-methyl-1-butanol concentration, mol/m^3
μ_{AA}	specific rate of amino acid uptake, hr^{-1}
μ_{AAl}	maximum velocity for AAl formation, hr^{-1}
μ_{EC}	maximum velocity for EC formation, hr^{-1}
μ_G	maximum velocity for glucose, hr^{-1}
μ_{i0}	Arrhenuis frequency factor, hr^{-1}
μ_I	specific rate of isoleucine uptake, hr^{-1}
μ_{IA}	specific rate of IA formation, hr^{-1}
μ_{IAc}	maximum velocity for IAc formation, hr^{-1}
μ_L	specific rate or leucine uptake, hr^{-1}
μ_M	maximum velocity for maltose, hr^{-1}
μ_N	maximum velocity for maltotriose, hr^{-1}
μ_S	specific uptake rate of S, hr^{-1}
μ_V	specific rate of valine uptake, hr^{-1}
μ_{VDK}	maximum velocity for VDK formation, hr^{-1}
μ_x	specific yeast growth rate, hr^{-1}
μ_1	specific glucose uptake rate, hr^{-1}
μ_2	specific maltose uptake rate, hr^{-1}
μ_3	specific maltotriose uptake rate, hr^{-1}
n	the biomass inhibition exponent
N	maltotriose concentration, mol/m^3
$[OA_j]$	oxo acid concentration, mol/m^3
$[OH_j]$	concentration of alcohol, mol/m^3
P	n-propanol concentration, mol/m^3
ρ	density of wort, kg/m^3
T	temperature, °C
t	time, hrs
T_c	effective bulk coolant temperature, °C
τ_d	first-order time constant for delay, hr
u	cooling control, J/m^3/hr/°C
V	valine concentration, mol/m^3
$[VDK]$	vicinal diketone concentration, mol/m^3
X	yeast concentration, mol/m^3
X_0	the initial biomass concentration, mol/m^3
$Y_{AA/X}$	yield coefficient, mols amino acid needed per mol biomass growth
Y_{AAl}	yield coefficient, mols AAl formed per mol sugar fermented
$Y_{AAl/X}$	yield coefficient, mols AAl formed per mol biomass growth
Y_{CG}	yield coefficient, mols C per mol G
Y_{CM}	yield coefficient, mols C per mol M
Y_{CN}	yield coefficient, mols C per mol N
$Y_{EA/S}$	ethyl acetate yield per mole sugar fermented
$Y_{EC/X}$	yield coefficient for ethyl caproate per mol biomass growth
Y_{EG}	yield coefficient, mols E per mol G
Y_{EM}	yield coefficient, mols E per mol M
Y_{EN}	yield coefficient, mols E per mol N
$Y_{IA/E}$	isoamyl alcohol yield, Ehrlich mechanism
$Y_{IA/S}$	isoamyl alcohol yield, synthetic pathway
Y_{IAc}	yield coefficient, mols isoamyl acetate produced per mol isoamyl alcohol formed
$Y_{IB/E}$	isobutyl alcohol yield, Ehrlich mechanism
$Y_{IB/S}$	isobutyl alcohol yield, synthetic pathway
Y_{IX}	yield coefficient, mols isoleucine needed per mol biomass growth
Y_{LX}	yield coefficient, mols leucine needed per mol biomass growth
$Y_{MB/E}$	2-methyl-1-butanol yield, Ehrlich mechanism
$Y_{MB/S}$	2-methyl-1-butanol yield, synthetic pathway
$Y_{OH/E}$	fusel alcohol yield, Ehrlich mechanism
$Y_{OH/S}$	fusel alcohol yield, synthetic pathway
$Y_{P/E}$	n-propanol yield from total transported amino acids
$Y_{P/F}$	n-propanol yield from fatty acid breakdown
$Y_{P/X}$	n-propanol yield from fatty acid breakdown based on yeast growth
$Y_{PI/E}$	n-propanol yield from transported isoleucine
$Y_{PI/S}$	n-propanol yield, synthetic pathway for isoleucine
$Y_{PV/E}$	n-propanol yield from transported valine
$Y_{PV/S}$	n-propanol yield, synthetic pathway for valine

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Y_{VDK} yield coefficient, mols *VDK* formed per mol biomass growth
Y_{VK} yield coefficient, mols valine needed per mol biomass growth
Y_{XG} yield coefficient, mols *X* per mol *G*
Y_{XM} yield coefficient, mols *X* per mol *M*
Y_{XN} yield coefficient, mols *X* per mol *N*

REFERENCES

1. Äyräpää, T. *Journal of the Institute of Brewing*, 1961, 67, 262.
2. Äyräpää, T. *Proceedings of the European Brewery Convention Congress, Brussels*, 1963, 276.
3. Äyräpää, T. & Palmqvist, U. *Journal of the Institute of Brewing*, 1970, 76, 144.
4. Dennis, J. E. Jr. and Schnabel, R. B. *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, Prentice-Hall, Englewood Cliffs, NJ, 1983.
5. Ehrlich, F. *Bericht daten chemische Gesellschaft*, 1906, 39, 4072.
6. Engasser, J. M., Marc, I., Moll, M. and Duteurtre, B. *Proceedings of the European Brewery Convention Congress, Copenhagen*, 1981, 579.
7. Gee, D. A. *"Modeling, Optimal Control, State Estimation and Parameter Identification Applied to a Batch Fermentation Process,"* Ph.D. Thesis, University of Colorado, 1990.
8. Gee, D. A. & Ramirez, W. F. *Biotechnology and Bioengineering*, 1988, 31, 224–234.
9. Gee, D. A. & Ramirez, W. F. *"On-line State Estimation and Parameter Identification for Batch Fermentation,"* submitted to *Biotechnology Progress*, 1994.
10. Geiger, E. & Piendl, A. *Brewers' Digest*, 1975, 50, 50.
11. Geiger, E. & Piendl, A. *Technical Quarterly of the Master Brewers Association of the Americas*, 1976, 13, 51.
12. Greenberg, D. M. *Metabolic Pathways*, Academic Press, NY, 1967.
13. Harding, S. A. & Kirsop, B. H. *Journal of the Institute of Brewing*, 1979, 85, 171.
14. Nordström, K. *Proceedings of the European Brewery Convention Congress Stockholm*, 1965, 195.
15. Taylor, G. T., Thurston, P. A. & Kirsop, B. H. *Journal of the Institute of Brewing*, 1979, 85, 219.
16. Thorne, R. S. W. *Journal of the Institute of Brewing*, 1937, 43, 288.

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